TRADEWISE
ENTITY–RELATIONSHIP DIAGRAM

As-built database reference · Reverse-engineered from the supplied Prisma schema and SQL migrations

Schema inventory	25 models · 11 enums · PostgreSQL
Relationships	38 declared Prisma foreign-key relations
Source of truth	schema.prisma + five SQL migrations
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How to read

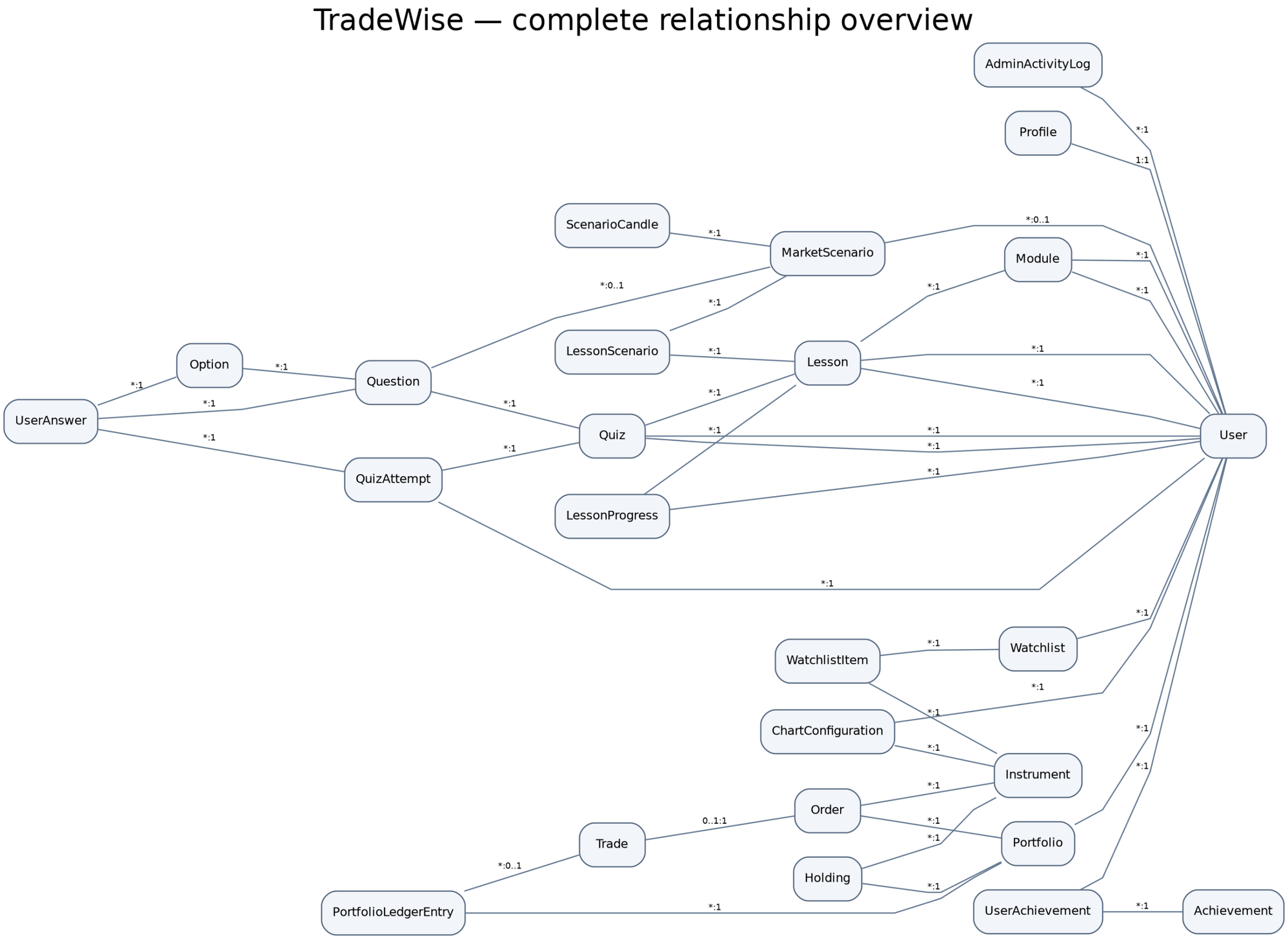
PK = primary key · FK = foreign key · UQ = unique field. Relationship labels show child-to-parent cardinality and the FK mapping. 0..1 indicates an optional relation. Junction models are shown as actual tables, including their own fields and composite keys.

Scope

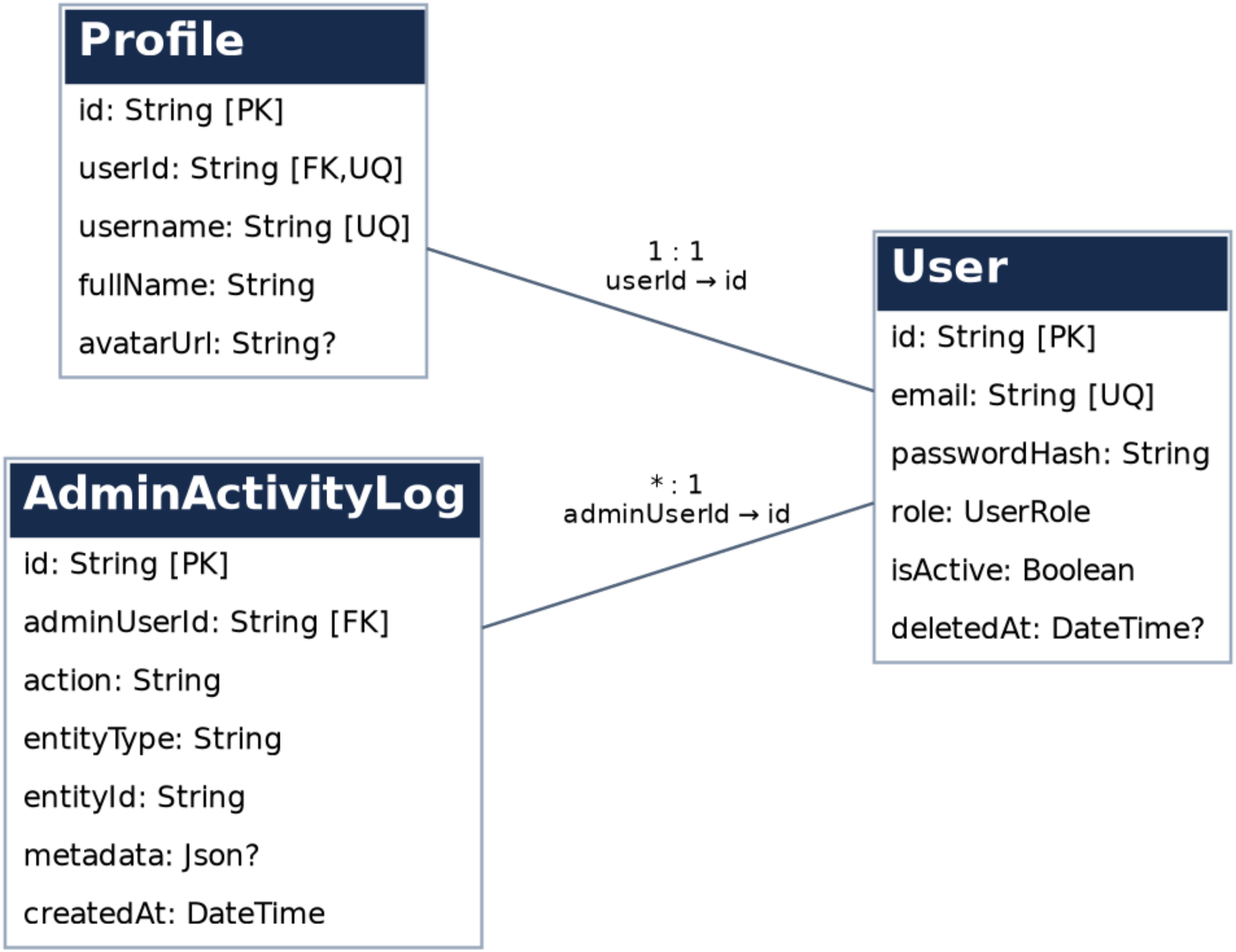
This is a structural, as-built ER reference. It includes only models and relations in the supplied schema. SQL-only CHECK constraints and triggers are summarized separately; they are not drawn as relationship edges. Enum types are listed on the final page.

Complete relationship overview

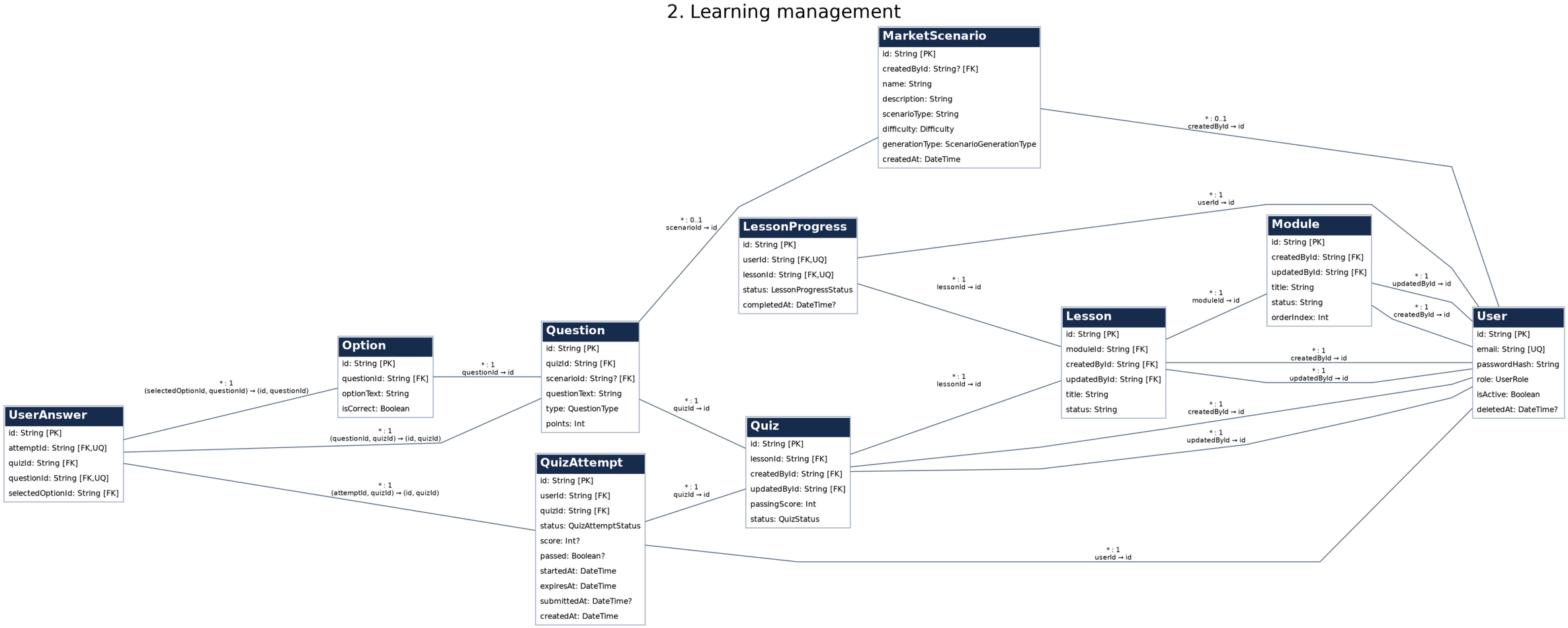
All 25 implemented models and declared foreign-key relationships. Use the domain pages for field-level detail.



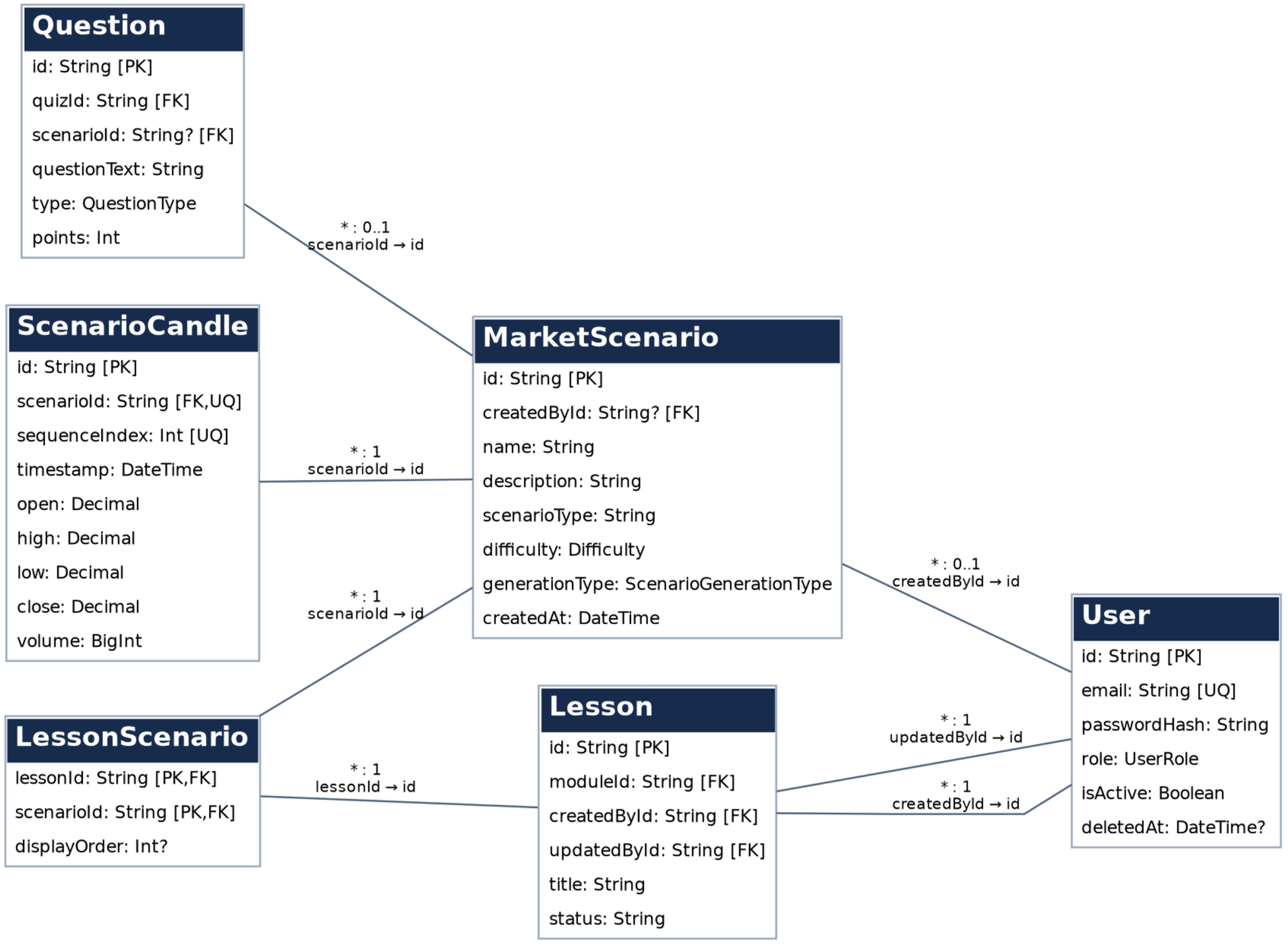
1. Identity & access



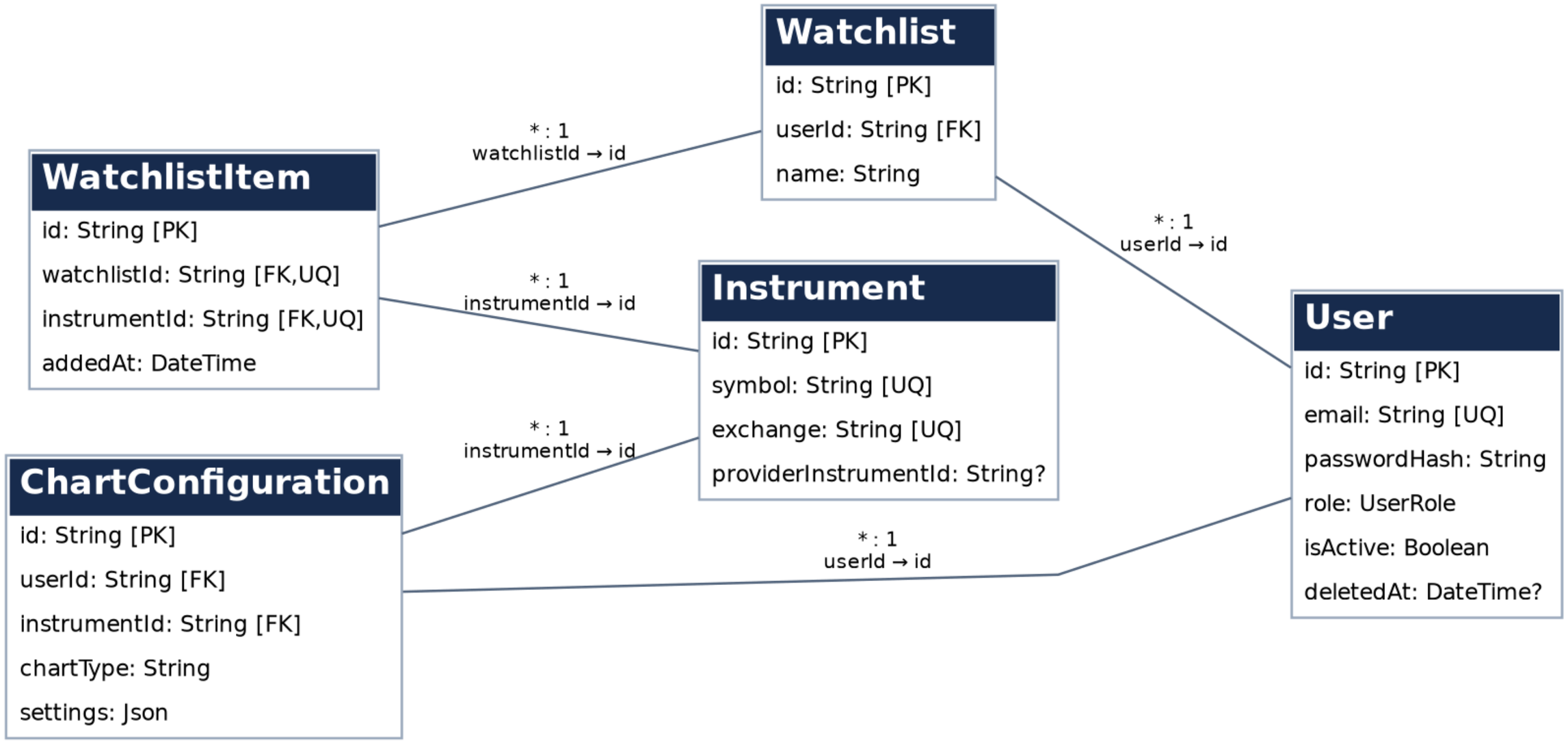
2. Learning management



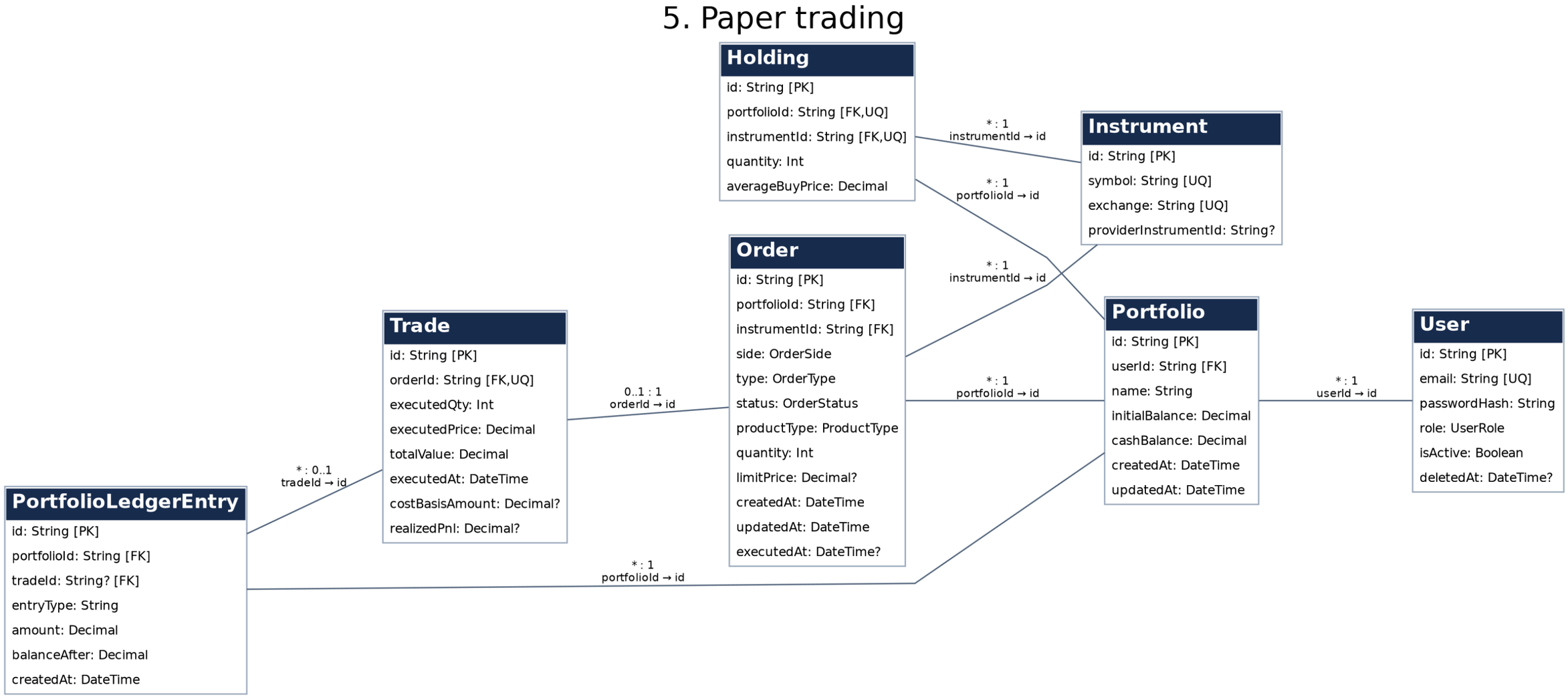
3. Scenarios & synthetic candles



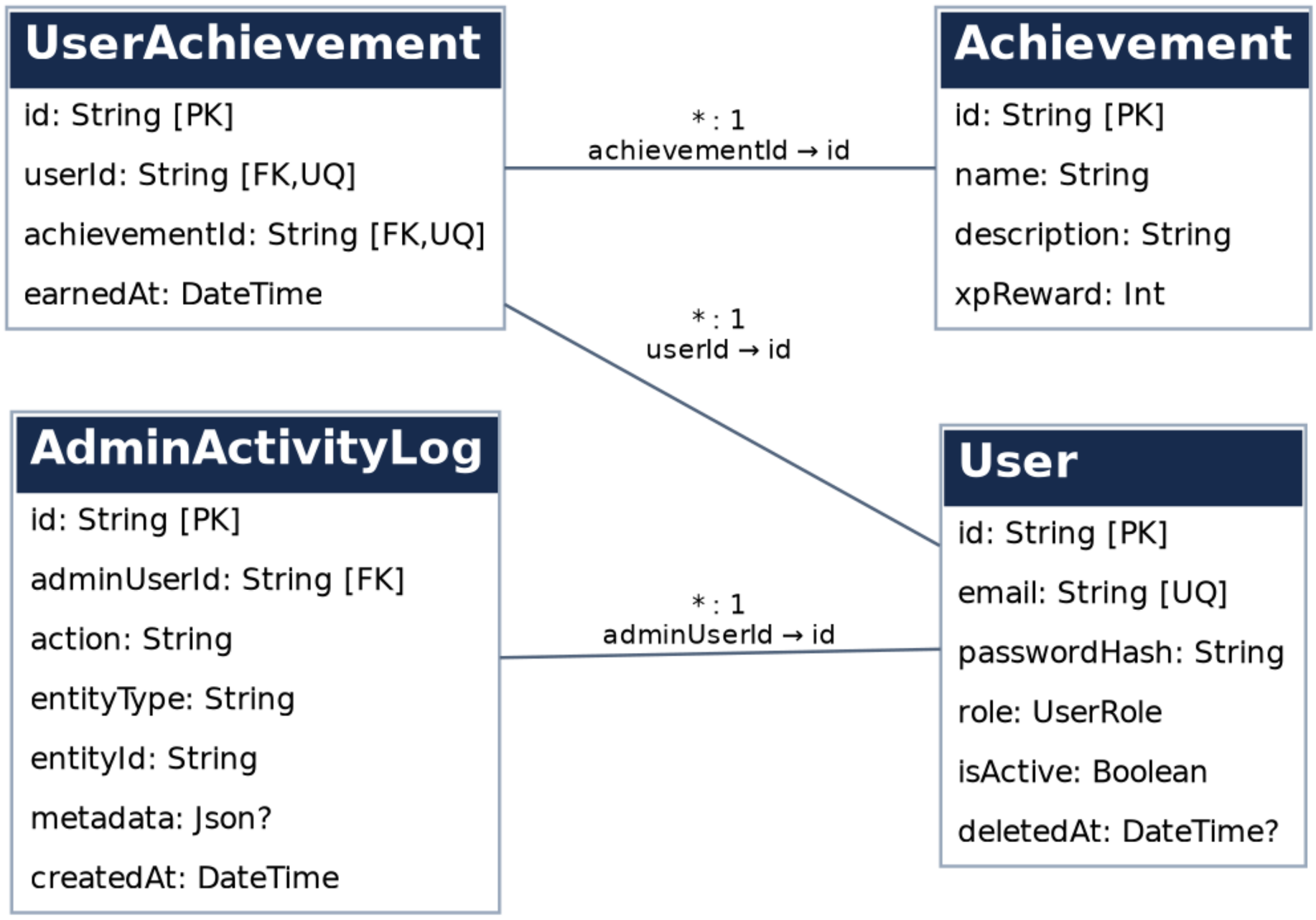
4. Instruments, watchlists & charts



5. Paper trading



6. Achievements & admin audit



Database rules beyond ER relationships

These rules affect real database behavior but are not relationship edges:

PostgreSQL CHECK constraints

Candle OHLC consistency and non-negative volume; positive order quantity; valid MARKET/LIMIT price combinations; positive executed trade quantity, price and value; positive holding quantity and average buy price.

Ledger immutability

SQL triggers reject UPDATE and DELETE on PortfolioLedgerEntry. Corrections are represented by compensating entries.

Deletion behavior

Explicit Prisma onDelete actions are defined in the schema. Relations without an explicit action use Prisma/database defaults; application-level lifecycle rules are separate.

Composite integrity

UserAnswer references QuizAttempt, Question and Option through composite keys containing quizId/questionId, preventing cross-quiz answer linkage.

Uniqueness

Examples: User.email; Profile.userId and username; Instrument(exchange, symbol); Trade.orderId; Portfolio+Instrument holdings; Watchlist+Instrument items; User+Lesson progress; User+Achievement.

Not represented as ER entities

Enums, service workflows, row locks, transactional behavior and application-only validation are not FK relationships.

Schema enums (11)

QuizStatus · UserRole · LessonProgressStatus · QuestionType · Difficulty · ScenarioGenerationType · OrderSide · OrderType · OrderStatus · ProductType · QuizAttemptStatus

Maintenance rule

Regenerate this document whenever schema.prisma or a database migration changes. Review the resulting PostgreSQL schema before treating a new revision as authoritative.